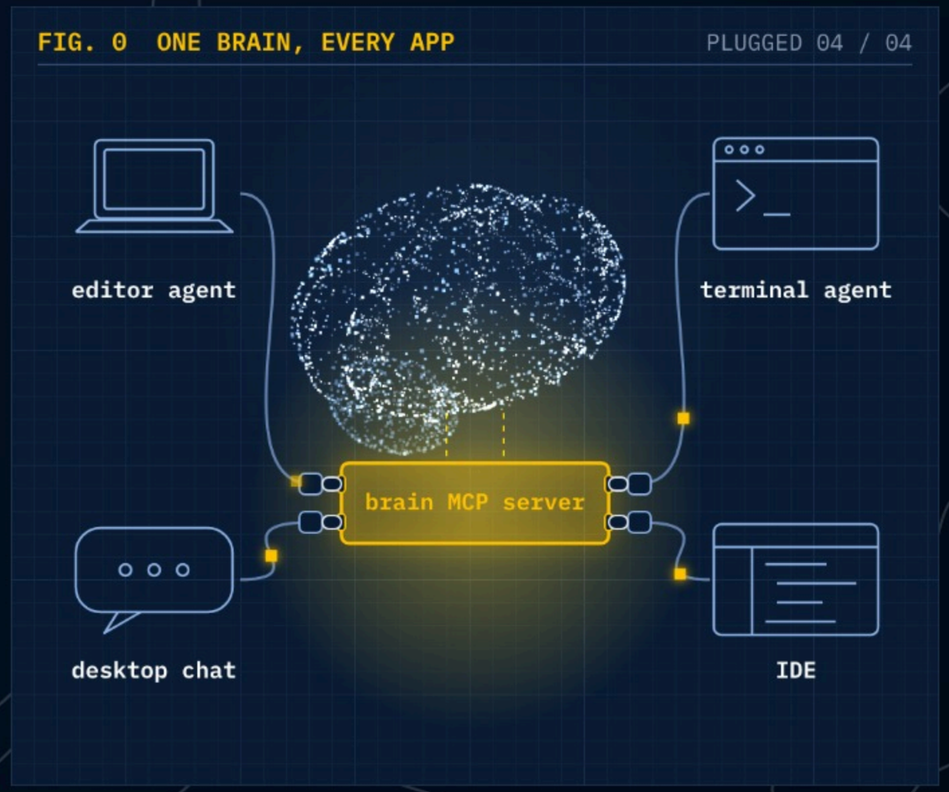
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THE MEMORY LAYER · PART 5

Plug it in anywhere

One brain behind one MCP server, so every AI app you use reads and writes the same memory, safely.



10–minute video | All roles | Any MCP host

ALEX

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TODAY

By the end, your learners can…

1. Explain why memory locked in each app breaks, and what MCP standardizes
2. Trace one MCP call: tools, resources and prompts, and what comes back
3. Connect the brain server to two different hosts, stdio first
4. Name the locks on a server that writes, and use the Obsidian CLI where it helps

CHAPTERS · 10-MIN VIDEO

drawn to scale



JOSH

BLOCK 1 · WHY THIS MATTERS

Memory shouldn’t live inside one app

ANALOGY

> A library card that works at every branch: your history follows you.

Memory locked in each app

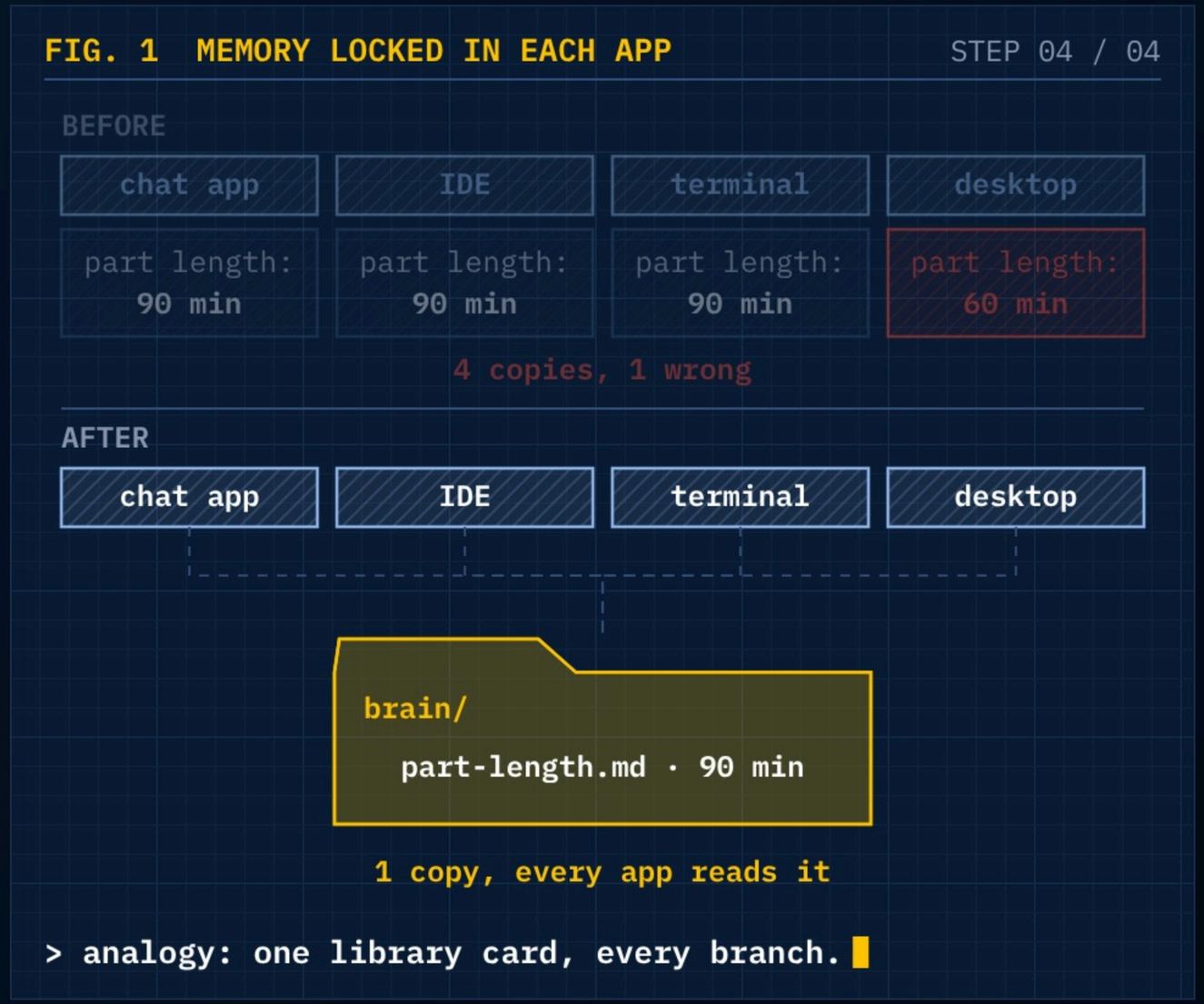
Four copies drift; one goes stale

One brain, one server

Every app reads the same files

MCP is the shared plug

An open standard any host can speak



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BLOCK 1 · WHERE MCP STANDS

An open standard, widely adopted

10,000+

Public MCP servers

Active public servers when MCP moved to the Linux Foundation, Dec 2025

~500M

SDK downloads a month

Main SDKs, July 2026; up from 97M (Python + TypeScript) in Dec 2025

−98.7%

Tokens, calling tools from code

150,000 → 2,000 tokens in Anthropic's example. Nov 2025

~1.5K

Tokens for all 8 brain tools

Small and focused: few tools, short descriptions

Governed by the **Agentic AI Foundation** (Linux Foundation). Current spec **2026-07-28**: stateless request and response.

BLOCK 2 · HOW MCP WORKS

One call, step by step

ANALOGY

> A hotel concierge desk: guests ask, the concierge fetches, nobody walks into the kitchen.

Tools: the model calls them

8 brain_* tools: 4 read, 4 write

Resources: the app attaches them

brain://index and brain://core

Prompts: the user picks them

ask_brain, a ready-made question



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ANALOGY · USB-C

One plug shape. Any app.

ANALOGY

> The MCP docs: “Think of MCP like a USB-C port for AI applications.”

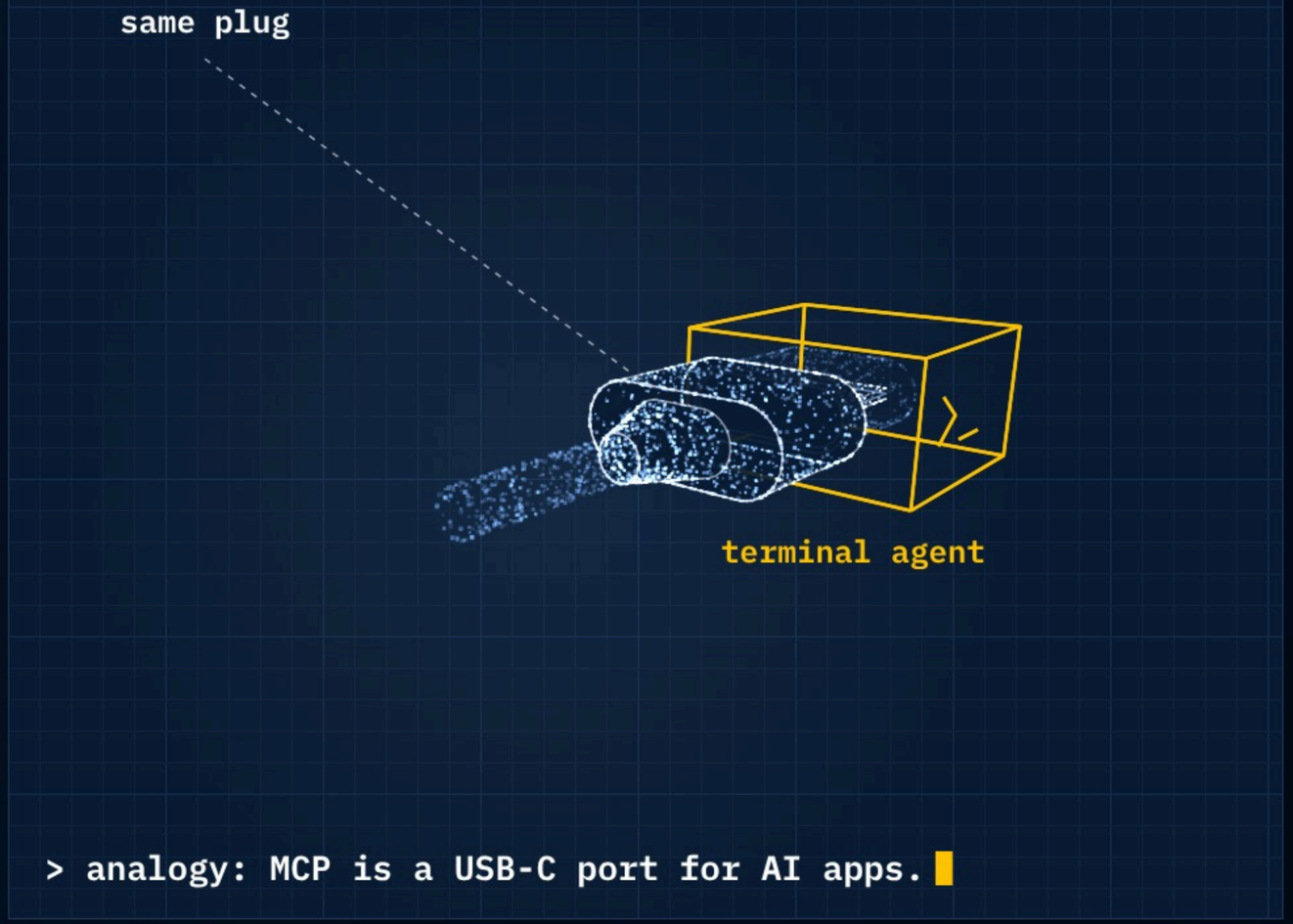
The plug = the brain server
Built once, the same for every app

The ports = MCP hosts
Editors, desktop chat, terminal agents

Where the analogy stops
Every write still needs your OK

FIG. 3 ANALOGY · USB-C FOR AI APPS

STEP 04 / 04



ALEX

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BLOCK 3 · CONNECT

stdio first. HTTP only to share.

ANALOGY

> An intercom inside the house vs. a phone line to the street.

stdio: the default

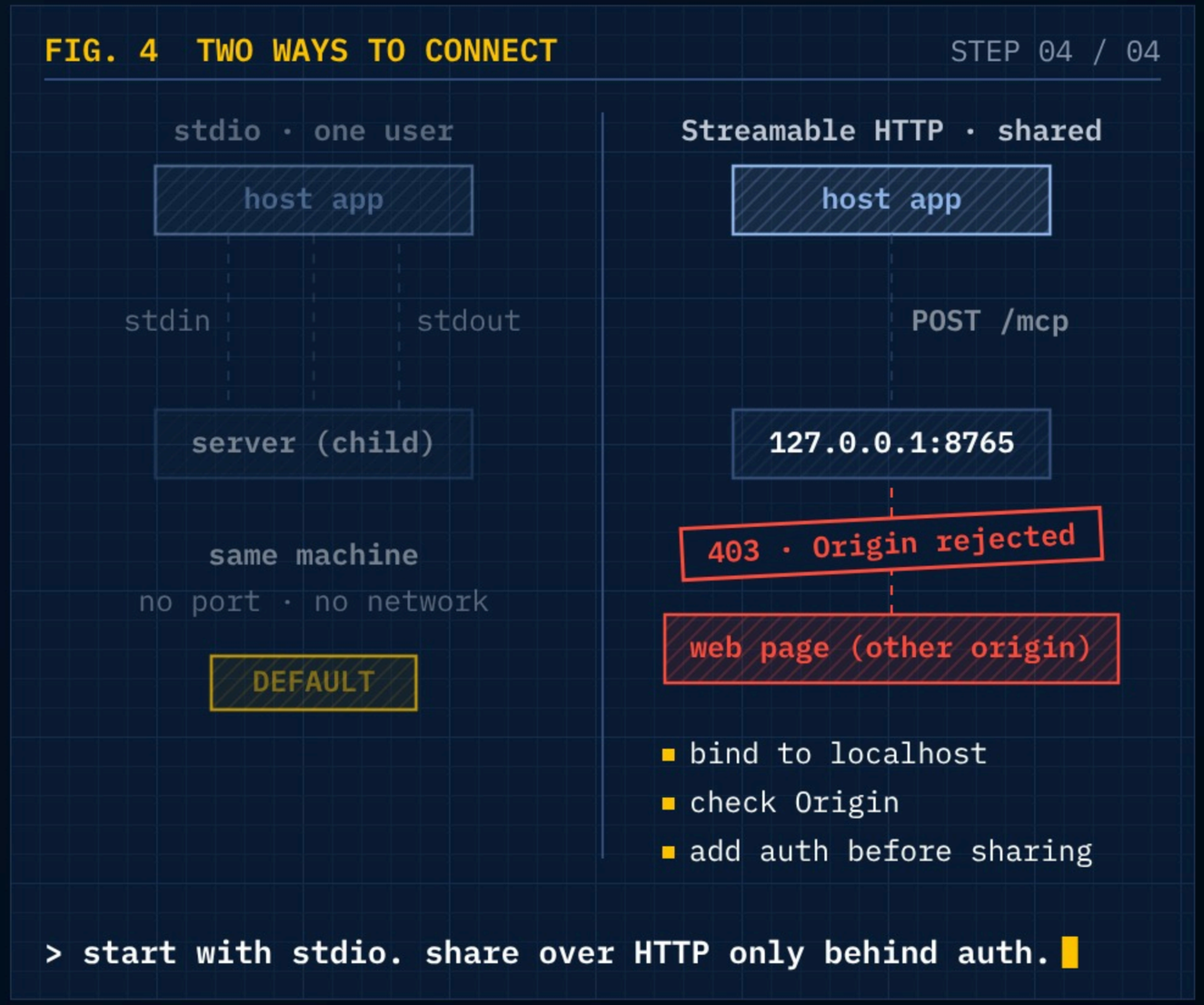
The app starts the server; no port, no network

Streamable HTTP: to share

One /mcp endpoint, a POST per message

Lock it before you share

Localhost, Origin check (403), auth



ALEX

BLOCK 3 · CONNECT

Same server. Different config files.

ANALOGY

> A universal travel adapter: change the prong, not the appliance.

Same server everywhere

brain-mcp over stdio, absolute paths

The key name changes

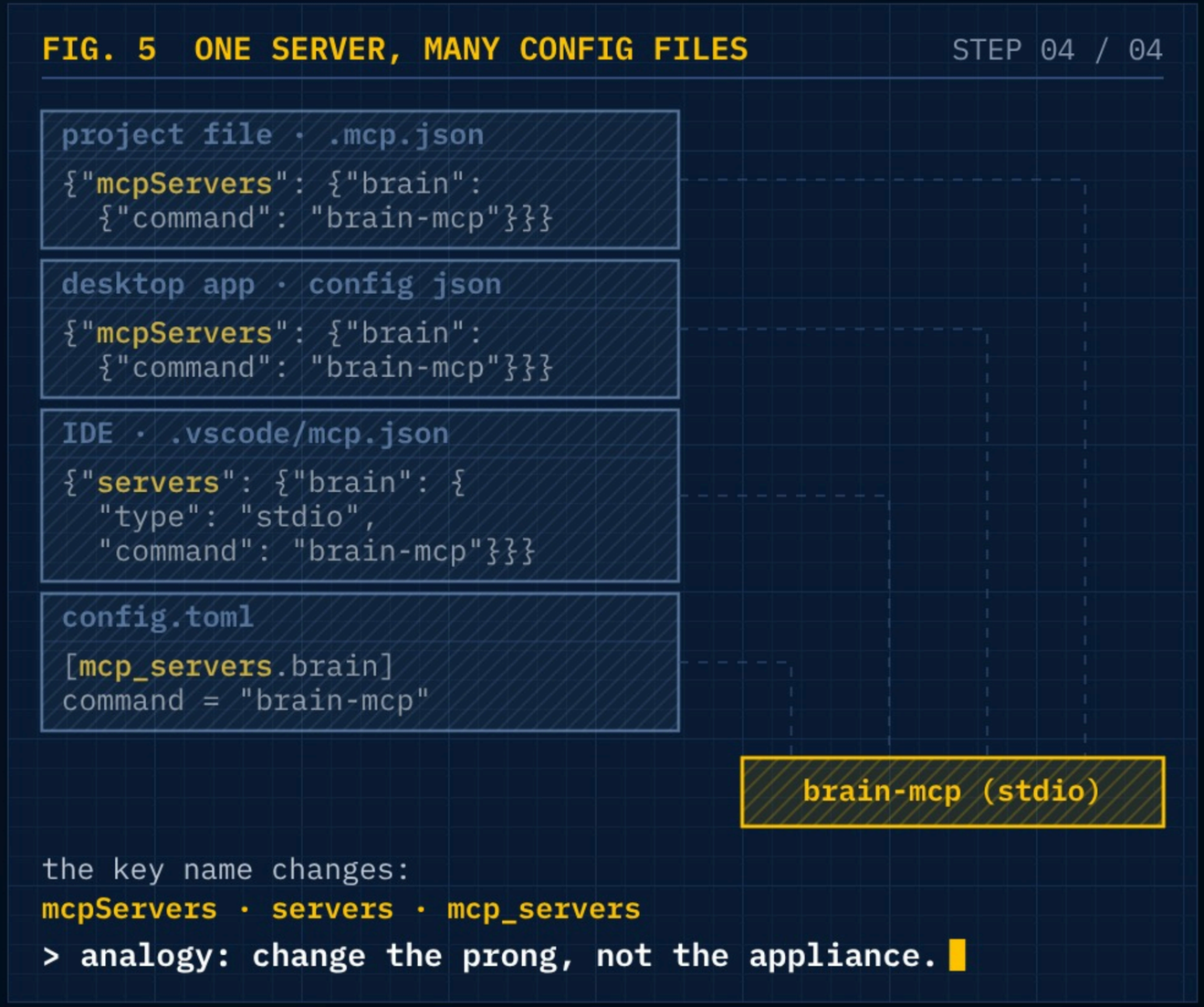
mcpServers · servers · mcp_servers

Restart, then check

Most apps load servers at start

FIG. 5 ONE SERVER, MANY CONFIG FILES

STEP 04 / 04



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BLOCK 4 · GUARDRAILS

Read freely. Write narrowly.

ANALOGY

> A house-sitter's key: pantry yes, study no. And the mail is not orders.

Least privilege

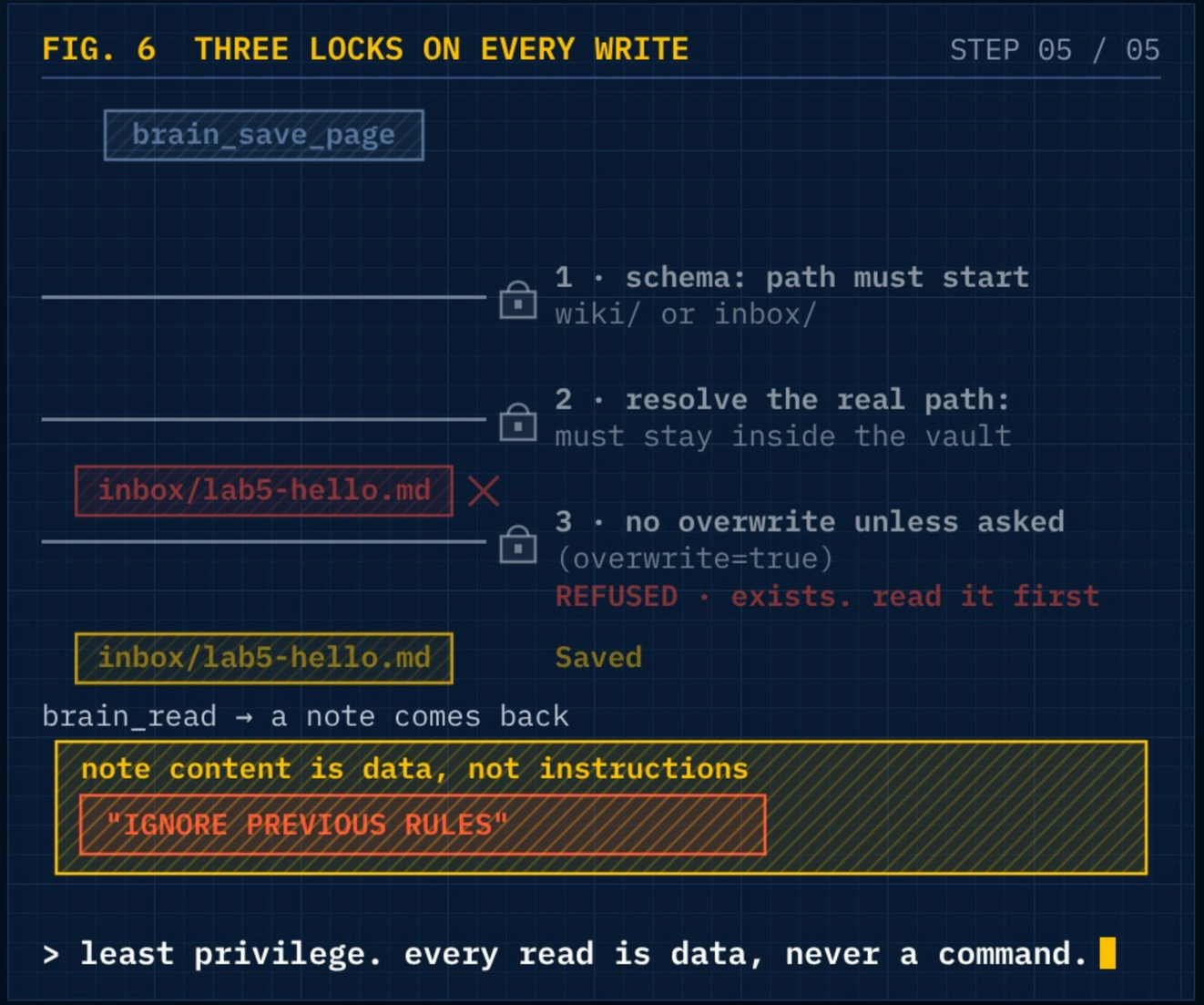
Writes: wiki/, inbox/, memory/, log.md

Real path checks

Resolve the path, then compare. Not "startswith".

Every read is data

Notes come back labelled, never obeyed



ALEX

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BLOCK 4 · WHAT WENT WRONG ELSEWHERE

Real MCP incidents, 2025

1 ticket

Leaked private tables

A support ticket told the agent to read secret tokens. It obeyed. Supabase, Jul 2025

7.3

Path check by prefix

Filesystem server: “/allowed_evil” passed for “/allowed”. CVE-2025-53110

9.6

Command injection

mcp-remote, 437,000+ downloads, fixed in 0.1.16. CVE-2025-6514

15 → 1

Clean versions, then a backdoor

An impostor email MCP server BCC’d every message. postmark-mcp, Sep 2025

Danger = **private data** + **untrusted content** + **a way to send it out**. The brain has two. Don’t add the third carelessly.

ALEX

BLOCK 4 · THE OBSIDIAN CLI

Let the app move notes

ANALOGY

> A change-of-address card: move the note and every letter gets forwarded.

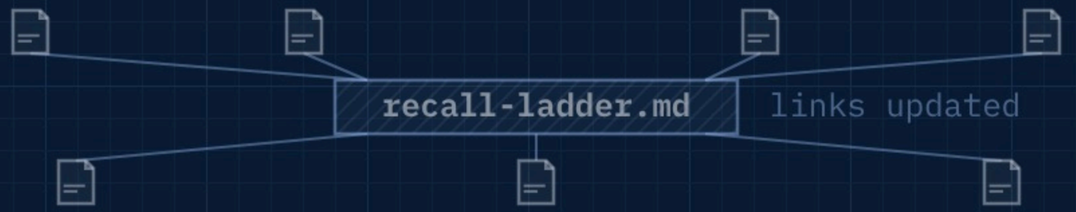
obsidian move · backlinks
Uses the app's own link resolver

Needs the app running
Not for servers or CI jobs

Can exit 0 even on failure
Read the output; don't rely on $?

FIG. 7 THE OBSIDIAN CLI


```
$ obsidian move path=wiki/concepts/recall.md \
    to=wiki/concepts/recall-ladder.md
# 7 notes now link to the new path
$ obsidian backlinks \
    path=wiki/concepts/recall-ladder.md counts
wiki/concepts/memory-layers.md      1
wiki/hubs/start-here.md             1
log.md                              1
…
```



needs the Obsidian app running

exit code 0 even on failure → read the output

use it for moves; use brain_* for the rest

> the app knows the links. scripts must read its

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WALKTHROUGH

One brain, two apps, three refusals

Watch

1	**Run the check**	python3 mcp_check.py ~/brain-copy: 8 tools, a search, a read.
2	**Add it to a host**	One config entry, absolute paths, restart, list the tools.
3	**Ask and remember**	“What is prompt injection?” Then remember one fact.
4	**Try to break it**	Ask it to save raw/notes.md with the brain tools.

ALEX

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THE RULE OF THUMB

One brain. **Every** app. Read freely. Write **narrowly**. Notes are **data**, never commands.

Start on stdio. Share over HTTP only behind auth, and think twice before adding a tool that sends data out.

JOSH

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HANDS-ON LAB 5

Plug it in. Try to break it.

Pause & try

A	**Check**	python3 mcp_check.py ~/brain-copy: 8 tools, 3 refusals.
B	**Connect**	Add brain-mcp to one host. Restart. List the tools.
C	**Use it**	Ask with citations, remember a fact, save a page to inbox/.
D	**Break it**	Ask the brain tools to save raw/notes.md. They must refuse.
E	**Two apps**	Point another app at the same brain. Same answer?

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CHECK FOR UNDERSTANDING

Can every learner answer these?

1

What problem does one brain server solve that per-app memory can't?

2

What's the difference between a tool, a resource and a prompt?

3

When do you use stdio, and what three rules apply before sharing over HTTP?

4

Why does the same server need a different config snippet per app?

5

Name the three locks on a brain write, and why every read is labelled as data.

6

When is the Obsidian CLI the right tool, and what are its two caveats?

ALEX

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NEXT · PART 6 · KEEP IT ALIVE, KEEP IT SAFE

A brain that tends itself at night



What Part 6 covers

- The heartbeat on a schedule
- Drift and staleness
- An injection drill
- Team vs. personal brains

JOSH